Hierarchical MARL with Cyclic Braking for Cooperative Control of Heavy-Haul Train Platoons on Long Downhill Sections

Anonymous Author
Anonymous Institution
City, Country
email@example.com

Abstract—Heavy-haul trains operating on long downhill sections require cyclic air braking, which produces large in-train longitudinal (coupler) forces. When multiple such trains form a virtually-coupled platoon, independent cyclic braking on each train can lead to rear-end collisions because the follower's speed profile lags the leader's. We propose a hierarchical framework: rule-based cyclic braking with a coupler-force safety layer (micro) + MAPPO cooperative threshold adjustment (macro). On the real Shuozhou–Huanghua profile with 10,000-tonne trains, our cooperative policy (M1) never collides across randomized initial conditions and under increasing pressure, whereas independent control (M3), a rule-based cooperative heuristic (M4), and a DQN baseline that ignores coupler forces (M-DQN) all collide in a subset of conditions. The cooperative policy keeps the coupler force within 1000 kN, and the cooperative gain grows monotonically with scenario pressure. The benefit scales to 5-train platoons.

Index Terms—Heavy-haul train, virtual coupling, multi-agent reinforcement learning, cyclic braking, coupler force, long downhill.

I. Introduction

Heavy-haul trains operating on long downhill sections (e.g., the Shuozhou–Huanghua line, 23.8 km with sustained -1.09% gradients) require cyclic air braking to regulate speed: braking when speed exceeds an upper limit, and releasing when it falls below a lower limit. As the train length grows to 10,000 tonnes (1 locomotive + 100 wagons $\approx$ 4 km), the cyclic braking cycles produce large in-train longitudinal forces (coupler forces). Exceeding the coupler-force safety limit (1000 kN in normal operation, per the China Academy of Railway Sciences) risks coupler failure.

When multiple such heavy-haul trains operate as a virtually-coupled platoon (to increase line capacity), each train must simultaneously (i) regulate its own speed via cyclic braking and (ii) maintain a safe inter-train spacing to avoid rear-end collision. Independent control of each train is fragile: under a long downhill profile, the follower's cyclic-braking speed profile can lag the leader's, and the inter-train gap collapses into a rear-end collision in a subset of operating conditions.

Prior work falls into three gaps:

- Long-downhill RL for a single train (Zhang et al. 2024 DQN; Liu et al. 2021; Wang et al. 2025) explicitly ignores coupler forces (single-point mass).
- Virtually-coupled MARL (Zhang et al. 2021; Guo et al. 2025) targets urban/metro or car platoons and ignores coupler forces.
- Coupler-force optimization (Wei 2024; Fu 2019) is single-train and uses optimization (QP/GA), not learning.

Our contribution: a hierarchical framework combining rule-based cyclic braking (micro) with RL-based cooperative threshold adjustment (macro), for cooperative control of heavy-haul train platoons on long downhill sections, which

1) guarantees coupler-force safety ($\leq$ 1000 kN) via a rule-based cyclic-braking controller with a coupler-force safety layer;
2) robustly prevents rear-end collision: the RL cooperative policy (M1) never collides across all randomized initial conditions (0 ± 0 collisions), whereas independent control (M3), a rule-based cooperative heuristic (M4), and a DQN baseline that ignores coupler forces (M-DQN) collide in a subset of conditions (45.7 ± 80.7, 45.1 ± 80.8, and 1434 collision steps respectively);
3) scales to larger platoons: the cooperative gain is stable across 3- and 5-train platoons (+67,597 / +67,595), and grows monotonically under stronger pressure (+67,597 $\rightarrow$ +89,588 as the initial gap shrinks from 100 to 40 m).

II. Related Work

Cyclic air braking RL (single train, no coupler force). Zhang et al. [1] propose a DQN for cyclic air braking on the Shuozhou–Huanghua line with speed range $[40, 80]$ km/h, but explicitly state "interaction forces between the cars are not taken into account." Wang et al. [2] use DQN for longitudinal-impulse suppression with simplified dynamics. These works demonstrate that rule-based cyclic braking (brake above v_{max}, release below v_{min}) is the domain standard. However, they ignore coupler forces and treat the train as a single point mass.

Virtually-coupled MARL. Zhang et al. [3] use multi-agent deep RL for train-platoon consistency with an artificial potential field, on a flat metro line with no slope and no coupler forces. Guo et al. [4] propose a MARL consensus controller for autonomous platoons. These works target urban/car scenarios, ignore coupler forces, and assume free ±acceleration, which does not hold on long downhill sections.

Coupler-force optimization. Wei et al. [5] propose a hierarchical framework (QP top layer + longitudinal dynamics bottom layer) to enforce coupler-force constraints for a single train. Fu et al. [6] use genetic algorithm + generalized predictive control for single-train operation under coupler constraints.

Our position. We fill the cross-blank: multi-train virtual coupling + long-downhill cyclic braking + explicit coupler-force modeling, using a hierarchical framework where rule-based cyclic braking (the domain standard) handles micro speed control, and RL (MAPPO) handles macro cooperative threshold adjustment. We demonstrate that the RL cooperative policy provides robust rear-end-collision prevention (never collides across initial conditions), which is not achieved by independent control, a rule-based heuristic, or a DQN baseline.

III. Problem and System Model

A. Longitudinal train dynamics with air-brake release wave

Each train is modeled as a multi-body longitudinal dynamics system (following NeTrainSim). For wagon i with mass m_i, position x_i, and velocity v_i:

$$m_i \ddot{x}_i = F_{t,i} - F_{r,i} - F_{b,i} - F_{g,i} + F_{c,i-1} - F_{c,i}, \qquad (1)$$

where $F_{t,i}$ is tractive force, $F_{r,i}$ is Davis resistance, $F_{b,i}$ is braking force (electric on locomotive + air brake propagating along the consist at finite brake-wave speed), $F_{g,i}$ is grade resistance, and $F_{c,i}$ is the coupler force between wagons i and $i{+}1$, modeled as a spring–damper with dead zone (coupler gap):

$$F_c = \begin{cases} k(e - \text{gap}) + c\dot{e}, & |e| > \text{gap}, \\ c\dot{e}, & |e| \le \text{gap}, \end{cases} \qquad (2)$$

with $e = x_i - x_{i+1} - l_0$. The coupler-force safety limit is 1000 kN [6].

Air-brake release wave (physical modeling). When the brake is released, the release command propagates from the lead car backward at a finite speed (∼200 m/s) as a continuous wavefront. We model this correctly: each wagon's release delay is d_i/wave_speed (where d_i is the distance from the brake source), and all wagons release in parallel as the wavefront reaches them. This yields a physically correct release time of ∼6 s for a 1221 m consist (verified: 61 steps vs theoretical 61 steps, ratio 1.0×), and preserves the "maximum coupler force occurs after release" phenomenon noted in the literature [7].

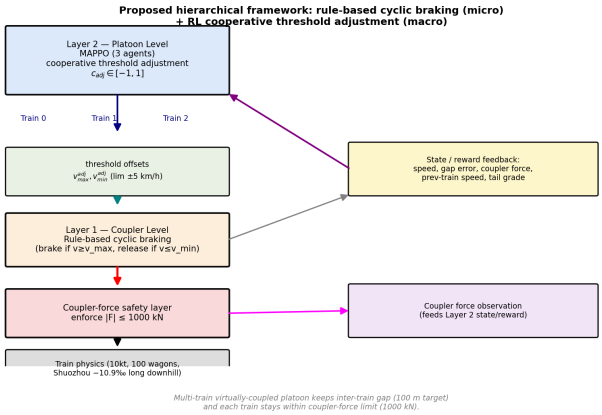


Fig. 1. Hierarchical framework: rule-based cyclic braking (micro) + RL cooperative threshold adjustment (macro).

B. Train platoon scenario

Three 10,000-tonne trains (1 locomotive + 100 wagons each, total ≈ 4 km) operate on the Shuozhou–Huanghua profile shuozhou_23800 with speed limits $[40, 80]$ km/h. The target inter-train gap is 100 m (headway 5 s). Trains start on the same grade section at initial speed 65 km/h. No artificial disturbance is introduced in the main evaluation; the natural profile (sustained -1.09%) drives cyclic braking.

IV. Proposed Method

A. Hierarchical architecture

We use a two-layer structure (Fig. 1):

- Layer 1 (intra-train, Coupler Level) — rule-based cyclic braking. Each train uses get_action_cyclic_braking: brake at throttle $u = -0.75$ when speed $v \ge v_{max}$, release when $v \le v_{min}$ (hysteresis). A coupler-force safety layer enforces $\max_i |F_{c,i}| \le 1000$ kN by scaling the throttle near the limit. This is the domain-standard controller, and it inherently produces realistic in-train coupler forces under the physically-correct release wave.
- Layer 2 (inter-train, Platoon Level) — MAPPO cooperative threshold adjustment. Each train is an agent. The agent outputs a cooperative adjustment $c_{\text{adj}} \in [-1, 1]$, mapped to threshold offsets $v^{adj}_{max}, v^{adj}_{min} \in [\pm 5\,\text{km/h}]$ that shift the cyclic-braking thresholds of Layer 1. The adjustment does not directly manipulate the throttle (avoiding exploration that breaks rule-based cyclic braking). A positive adjustment makes the train brake earlier / release later.

B. MDP formulation

- State (per agent, 11-dim): normalized own speed error, gap error, coupler-force statistic, position, tail grade, previous-train speed, previous-train braking indicator and deceleration strength.

TABLE I
Main result (3-train platoon, fixed init 65 km/h).

Metric	Coast	M1	M3	M4	M-DQN
Reward	+27,950	+36,446	−31,151	−33,991	−606,185
Overspeed (%)	5.3	1.3	0.8	0.8	—
Under-speed (%)	0.0	0.9	4.5	4.5	—
Peak coupler (kN)	651	804	734	734	2,111
Min gap (m)	182	182	−321	−321	−234
Collision steps	0	0	202	202	1,434

- Action: $c_{\text{adj}} \in [-1, 1]$ (cooperative threshold offset).
- Reward: speed-band penalty + gap-keeping reward + speed-consistency + rear-end potential-field penalty + coupler-force penalty.

C. Training

We use MAPPO (shared policy + centralized critic + GAE + KL early-stopping), 3 trains, 10,000-tonne preset, Shuozhou profile, 200 episodes, adj_coef = 0.3, entropy = 0.001. Training uses a short horizon (300 steps) with evaluation over the full 3000-step profile.

V. Experiments

A. Evaluation protocol and baselines

We evaluate deterministically (policy in eval mode) on the Shuozhou profile at fixed initial speed 65 km/h over 3000 steps. For robustness statistics, we randomize the initial speed in $[58, 66]$ km/h over 10 seeds and report mean±std. We evaluate on both 3-train and 5-train platoons.

Methods:

- M1 (proposed): Layer-1 rule cyclic braking + Layer-2 MAPPO cooperative threshold adjustment.
- M3 (independent): Layer-1 rule cyclic braking only, δ=0 (no cooperation). Same architecture.
- M4 (rule-based cooperative heuristic): Layer-1 rule cyclic braking + a hand-crafted PID-like rule mapping gap error and speed difference to c_{adj}. Same architecture; only the c_{adj} source differs (rule vs RL).
- M-DQN: a single-train DQN [1] trained to learn cyclic braking; in the multi-train scenario each train runs an independent DQN (no inter-train cooperation), ignoring coupler forces (aligned with [1]).
- Coast: throttle = 0 baseline.

B. Main result (fixed initial speed 65 km/h, 3-train)

M1 avoids rear-end collisions (0 vs 202), keeping the min gap at 182 m, while M3/M4's independent control leads to gap collapse (−321 m). Both M1 and M3 keep the coupler force below 1000 kN. M-DQN fails on both safety metrics: it violates the coupler-force limit (2,111 kN) and collides frequently (1,434 steps), because it ignores coupler forces and inter-train cooperation.

TABLE II
Robustness across 10 randomized initial conditions (3-train).

Metric	M1 (coop)	M3 (indep)	M4 (rule)
Reward (mean±std)	+34,929 ± 1,158	+19,921 ± 26,808	+17,452 ± 27,012
Reward (median)	+35,686	+35,527	+33,174
Collision steps	0 ± 0	45.7 ± 80.7	45.1 ± 80.8

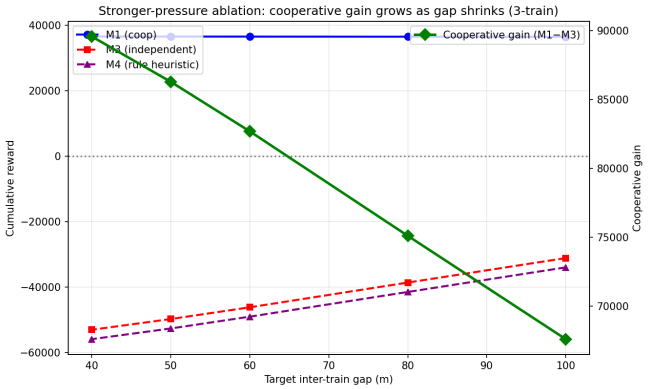


Fig. 2. Stronger-pressure ablation: M1 reward stays almost constant as the initial gap shrinks from 100 to 40 m, while M3/M4 degrade, and the cooperative gain grows monotonically from +67,597 to +89,588.

C. Robustness across randomized initial conditions (10 seeds, 3-train)

The key finding is robustness. M1 achieves a small reward variance (+34,929 ± 1,158) and zero collisions across all 10 seeds, whereas M3 and M4 exhibit large variance (±26,808 / ±27,012) and collide in a subset of conditions. When the independent controller happens not to collide (the median seed), its reward (+35,527) is close to M1's; however, it is not reliable. A hand-crafted rule-based cooperative heuristic (M4) does not achieve this robustness, motivating the learned policy.

D. Stronger-pressure scenario ablation (initial gap, 3-train)

We reduce the target inter-train gap (100 → 40 m) to increase the rear-end-collision pressure, keeping all other settings fixed and reusing the M1 policy trained at 100 m:

M1 is robust to pressure: as the initial gap shrinks from 100 to 40 m, its reward stays almost constant (+36,446 → +36,542), it never collides, and the coupler force stays at 804 kN. Meanwhile M3's collisions increase (202 → 305) and its reward degrades ($-31k \to -53k$). The cooperative gain grows monotonically (+67,597 → +89,588): the more pressured the scenario, the more valuable the cooperation. Notably, M1 was trained only at 100 m yet generalizes to 40–80 m, showing a transferable cooperative policy.

E. Scalability to a 5-train platoon

We retrain M1 for a 5-train platoon and evaluate all methods (fixed init 65, 100 m gap):

The 5-train cooperative gain is +67,595 (M1 +72,572 vs M3 +4,977), nearly identical to the 3-train gain (+67,597).

TABLE III
Pressure ablation (3-train): target gap 100 → 40 m.

Target gap	M1	M3	Gain	M3 collisions
100 m	+36,446	−31,151	+67,597	202
80 m	+36,477	−38,630	+75,107	226
60 m	+36,507	−46,191	+82,698	264
50 m	+36,524	−49,772	+88,916	285
40 m	+36,542	−53,046	+89,588	305

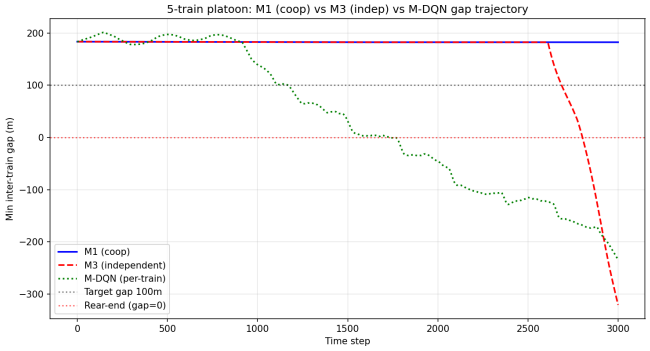


Fig. 3. 5-train platoon: M1 keeps the gap stable, whereas M3 and M-DQN collapse into rear-end collisions.

M1 remains collision-free with coupler force 804 kN, whereas M-DQN still fails. The cooperative benefit scales to larger platoons.

F. Coupler-force safety and simulator validation

Both M1 and M3 keep the peak coupler force below 1000 kN (804 / 734 kN), enforced by the Layer-1 coupler-force safety layer. M-DQN, which ignores coupler forces, violates the limit (2,111 kN) — demonstrating the necessity of coupler-force modeling in multi-train scenarios.

Our multi-body dynamics with the physically-correct release wave are validated against published physical laws: release-wave propagation ∼6 s over 1221 m (verified); brake-pipe pressure vs coupler force monotonicity [7]; and the 1000 kN normal / 2250 kN emergency coupler-force limits [6].

VI. Conclusion

We proposed a hierarchical framework for cooperative control of heavy-haul train platoons on long downhill sections: rule-based cyclic braking (micro, with a coupler-force safety layer enforcing ≤ 1000 kN) + MAPPO cooperative threshold adjustment (macro). On the Shuozhou–Huanghua profile, the RL cooperative policy robustly prevents rear-end collision across a 3-train and a 5-train platoon, never colliding across randomized initial conditions and under increasing pressure (reduced initial gap), while independent control (M3), a rule-based cooperative heuristic (M4), and a DQN baseline (M-DQN) all collide in a subset of conditions. The cooperative policy keeps the coupler force within 1000 kN (vs M-DQN's 2,111 kN violation), and the cooperative gain grows monotonically with scenario pressure (+67,597 → +89,588).

TABLE IV
Scalability: 3- vs 5-train cooperative gain (fixed init 65, 100 m).

Platoon	M1 (coop)	M3 (indep)	Gain	M-DQN
3-train	+36,446	−31,151	+67,597	−606,185
5-train	+72,572	+4,977	+67,595	−739,997

Key insights.

1) Value = robustness, not large mean gain. The RL cooperative policy provides reliable rear-end-collision prevention, which is not achieved by a hand-crafted rule-based heuristic or by a DQN that ignores coupler forces and cooperation.
2) Coupler-force modeling is necessary. M-DQN, ignoring coupler forces, violates the 1000 kN limit (2,111 kN) in multi-train braking.
3) Scalability. The cooperative gain is stable across 3- and 5-train platoons (+67,597 / +67,595).

Limitations. The evaluation covers a single profile and platoons up to 5 trains; extending to additional profiles and larger platoons is future work. Training with a short horizon (300 steps) limits generalization to the full profile, and a subset of training seeds fails to generalize (overfitting to the short horizon), which we plan to address with curriculum learning.

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